

HDOC Day – 15

Prepare the algorithm and test case table. Then write, compile, execute, and test the C program against the prepared use cases.

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

1. **Check wheather all its digits are unique.**
 - Example 1: Number: 12345
Output: **All digits not even**
 - Example 2: Number: 24680
Output: **All digits are even**
2. **Check whether all digits are prime.**
 - Example 1: Number: 2357
Output: **All digits are prime**
 - Example 2: Number: 2379
Output: **All digits are not prime**

Aim:

To write a menu-driven C program using **switch-case** to input a positive integer and perform the following operations according to the user's choice:

1. Check whether all digits of the number are even.
2. Check whether all digits of the number are prime.

Algorithm:

1. Start.
2. Input a positive integer n.
3. Display the menu:
 - 1 .Check whether all digits are even.
 - 2. Check whether all digits are prime.
4. Enter the user's choice.
5. Use switch-case for the selected choice.
6. For **Case 1**, check each digit of the number:
 - If every digit is divisible by 2, print **"All digits are even"**.
 - Otherwise, print **All digits are not even .**
7. For **Case 2**, check each digit:
 - If every digit is one of 2, 3, 5, 7, print **"All digits are prime"**.

- Otherwise, print **“All digits are not prime”**.
8. For any other choice, print **“Invalid choice”**.
 9. Stop.

Test Case Number Choice Expected Output

1	24680	1	All digits are even
2	12345	1	All digits are not even
3	2357	2	All digits are prime
4	2379	2	All digits are not prime
5	2468	3	Invalid choice

>Write a code C.

```

1#include <stdio.h>
2
3int main()
4{
5    int n, choice, digit, temp, flag = 1;
6
7    printf("Enter a positive integer: ");
8    scanf("%d", &n);
9
10   printf("\n1. Check whether all digits are even");
11   printf("\n2. Check whether all digits are prime");
12   printf("\nEnter your choice: ");
13   scanf("%d", &choice);
14
15   temp = n;
16
17   switch(choice)
18   {
19       case 1:
20           while(temp > 0)
21           {
22               digit = temp % 10;
23               if(digit % 2 != 0)
24               {
25                   flag = 0;
26                   break;
27               }
28               temp = temp / 10;
29           }
30           if(flag == 1)
31               printf("All digits are even");
32           else

```

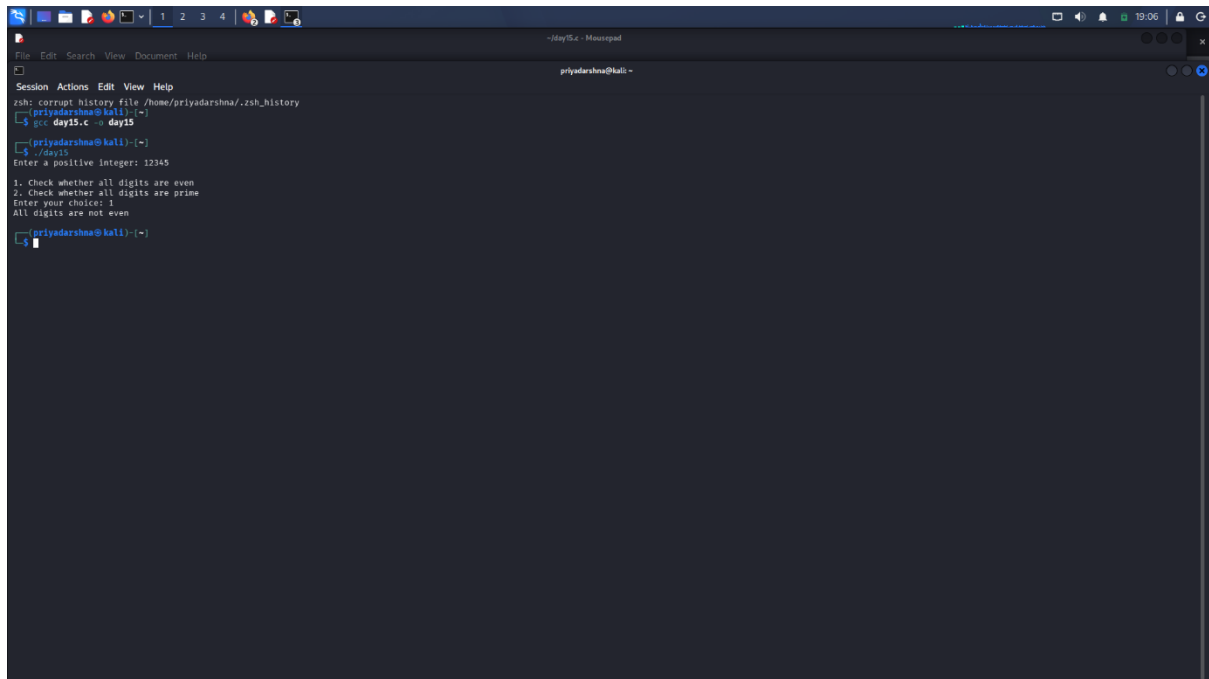
```
32
33     if(flag == 1)
34         printf("All digits are even");
35     else
36         printf("All digits are not even");
37     break;
38
39     case 2:
40         while(temp > 0)
41         {
42             digit = temp % 10;
43
44             if(digit != 2 && digit != 3 &&
45                digit != 5 && digit != 7)
46             {
47                 flag = 0;
48                 break;
49             }
50
51             temp = temp / 10;
52         }
53
54         if(flag == 1)
55             printf("All digits are prime");
56         else
57             printf("All digits are not prime");
58         break;
59
60     default:
61         printf("Invalid choice");
62 }
63
64 return 0;
65 }
66
```

>Test case 1:

```
File Edit Search View Document Help
--day15.c - Mousepad

Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
[priyadarshna@kali:~]$ gcc day15.c -o day15
[priyadarshna@kali:~]$ ./day15
Enter a positive Integer: 24680
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 1
All digits are even
[priyadarshna@kali:~]$
```

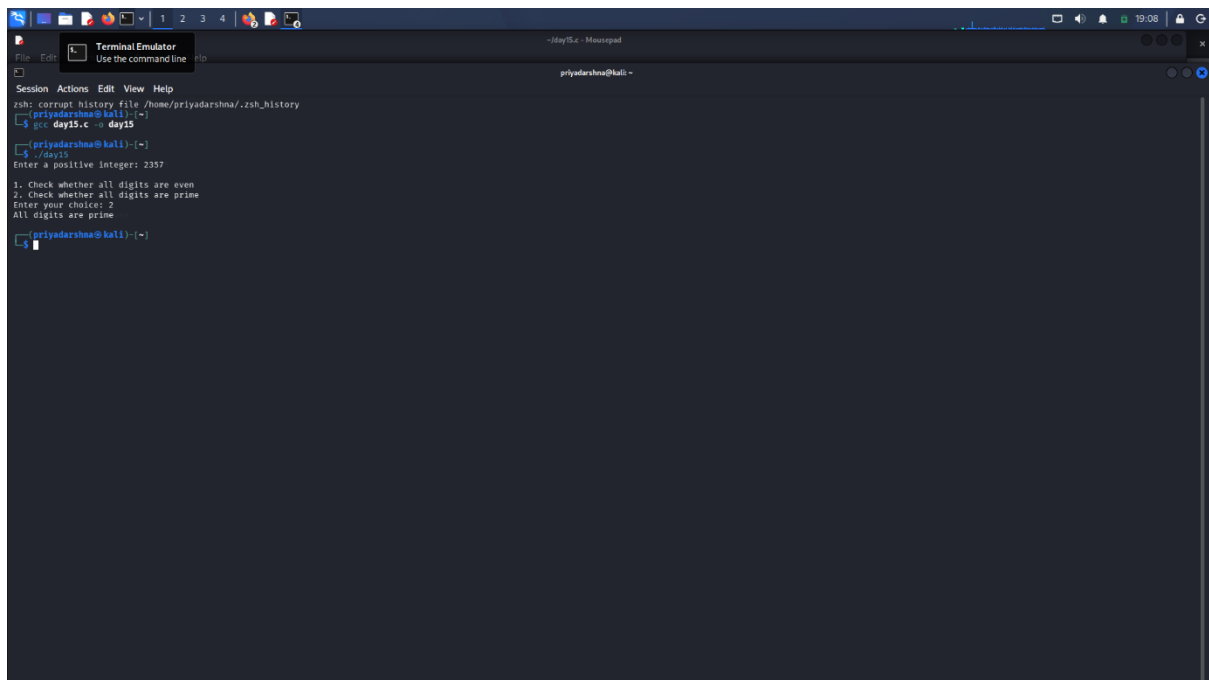
>Test case 2:



The screenshot shows a terminal window with the following content:

```
zsh: corrupt history file /home/priyadarshna/.zsh_history
[priyadarshna@kali]~$ gcc day15.c -o day15
[priyadarshna@kali]~$ ./day15
Enter a positive integer: 12345
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 1
All digits are not even
[priyadarshna@kali]~$
```

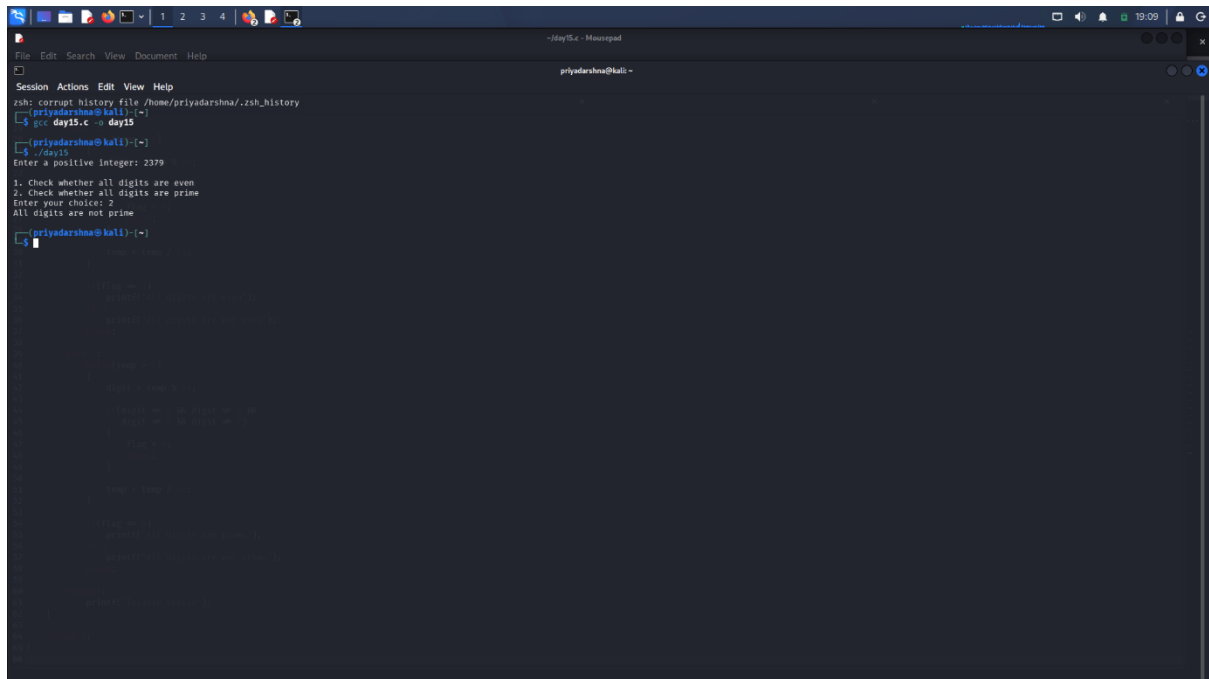
>Test case 3:



The screenshot shows a terminal window with the following content:

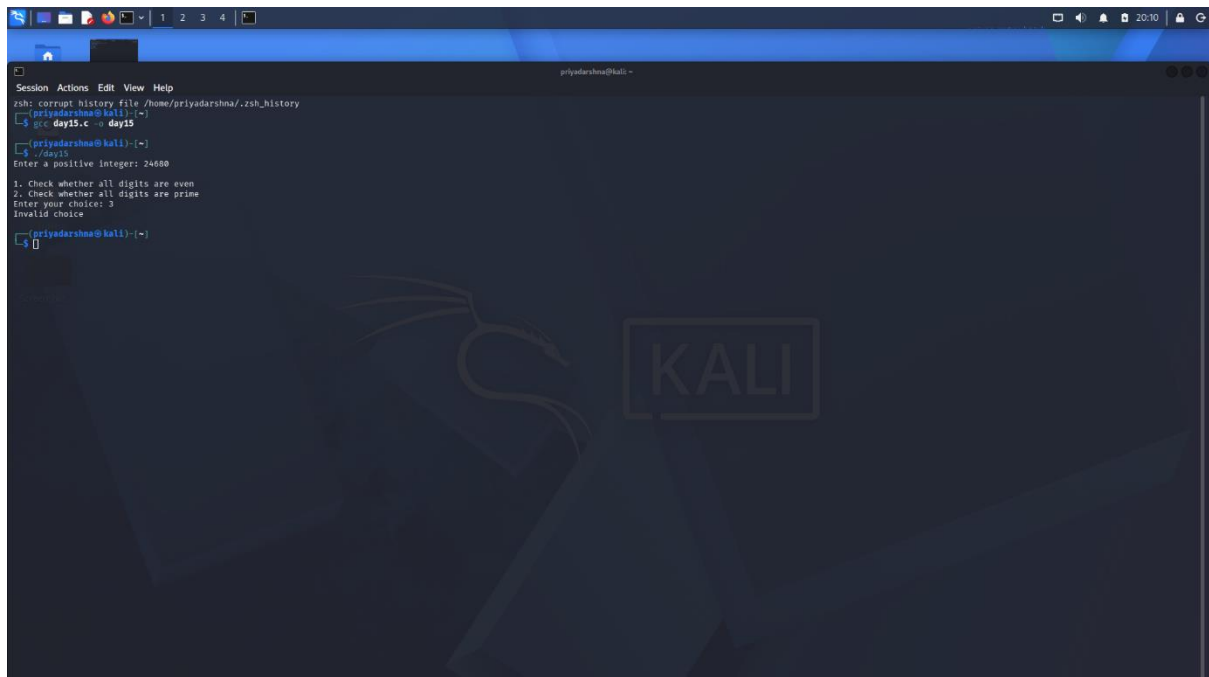
```
zsh: corrupt history file /home/priyadarshna/.zsh_history
[priyadarshna@kali]~$ gcc day15.c -o day15
[priyadarshna@kali]~$ ./day15
Enter a positive integer: 2357
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 2
All digits are prime
[priyadarshna@kali]~$
```

>Test case 4:



```
File Edit Search View Document Help
--day15.c - Mousetpad
Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
(priyadarshna@kali)~$ gcc day15.c -o day15
(priyadarshna@kali)~$ ./day15
Enter a positive integer: 2379
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 2
All digits are not prime
(priyadarshna@kali)~$
```

>Test case 5:



```
File Edit Search View Document Help
--day15.c - Mousetpad
Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
(priyadarshna@kali)~$ gcc day15.c -o day15
(priyadarshna@kali)~$ ./day15
Enter a positive integer: 24680
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 3
Invalid choice
(priyadarshna@kali)~$
```

Result:

The menu-driven C program using switch-case was successfully written, compiled, and executed. The program correctly checks whether:

All digits of the given number are even, or All digits of the given number are prime.

The program was tested successfully with the prepared test cases.